PHASE 3 OF 8

Requirements & Environment Setup

Everything you need before the first run: system prerequisites, Python packages and a Gemini API key.

System | Packages | API Key | Checklist

3.1 System Requirements

Requirement	Detail
Python	3.10 or newer
Disk space	~2–3 GB free (the local explanation model is a few hundred MB and downloads on first use)
API access	A Google Gemini API key from Google AI Studio
Internet	Needed only for Gemini-powered features: Q&A, Summarize, Quiz, Learning Path
Browser	Any modern browser able to run JavaScript and `fetch()`

3.2 Python Packages

All dependencies are listed in `requirements.txt`:

Package	Purpose
fastapi	Web framework that defines and serves the API routes
uvicorn[standard]	ASGI server that actually runs the FastAPI app
jinja2	Template engine used to render index.html
python-multipart	Enables form-data parsing for FastAPI requests
google-genai	Official client library for calling the Gemini API
transformers	Hugging Face library used to load the local explanation model
torch	Deep learning engine that transformers runs on
python-dotenv	Loads GEMINI_API_KEY from a local .env file

Tip

Install into a fresh virtual environment. `torch` is the largest dependency, so the first `pip install` can take several minutes on a slow connection.

3.3 Pre-Flight Checklist

	Item	How to verify
☐	Python 3.10+ installed	`python --version`
☐	Gemini API key created in Google AI Studio	Key visible in your AI Studio dashboard
☐	2–3 GB free disk space	Check your drive's free space
☐	`templates/index.html` and `static/style.css` exist	List the project folder
☐	Internet available for first run	Needed to download the local model weights

3.4 Step-by-Step Installation

Step 1 — Create a virtual environment

```
python -m venv venv
source venv/bin/activate      # Windows: venv\Scripts\activate
```

Step 2 — Install dependencies

```
pip install -r requirements.txt
```

Step 3 — Add your Gemini API key

Create a file named `.env` in the project root:

```
GEMINI_API_KEY=your_actual_key_here
```

Step 4 — Run the server

```
uvicorn main:app --reload
```

Step 5 — Open the app

Visit `http://127.0.0.1:8000` in your browser. The landing page should show the Q&A, Explain and Summarize forms.

> **First Explain request is slow — that is expected**
> The first call triggers a download of `MBZUAI/LaMini-Flan-T5-783M` (a few hundred MB). After that it is cached locally and loads quickly.

3.5 Checking Which Gemini Models You Can Use

If a Gemini-powered feature errors with a "model not found" style message, run `check_models.py`. It lists every model your API key is allowed to call, so you can confirm that the model name used in the code (currently `gemini-3.6-flash`) is available to your key.

```
python check_models.py
```

3.6 Configuration Files

.env.example (safe to commit)

Adding a checked-in template tells new contributors exactly which variable to set:

```
# .env.example
# Copy this file to .env and fill in your real key.
GEMINI_API_KEY=your_actual_key_here
```

.gitignore (keeps secrets out of version control)

```
.env
venv/
__pycache__/
*.pyc
.DS_Store
```

3.7 Common Setup Pitfalls

Symptom	Likely cause	Fix
"GEMINI_API_KEY is not set"	.env missing, misnamed or in the wrong folder	Name it exactly `.env` and place it next to `main.py`
`uvicorn` not found	Virtual environment not activated	Activate `venv`, then retry
Template or static error	templates/ or static/ folder missing	Create the folders and add the files
Slow or stalled first Explain	Model weights downloading	Wait for the download; later calls are fast

3.8 Verifying Your Installation

Check	Command / action	Expected result
Python version	`python --version`	3.10 or newer
Packages installed	`pip list`	fastapi, uvicorn, transformers, torch and google-genai listed
Key is readable	Open the app and ask a Q&A question	An answer, not an "API key not set" message
Server reachable	Visit `http://127.0.0.1:8000`	The EduGenie landing page loads